



Protocol name: RP01 Preparation of aqueous uranyl acetate

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Keywords: Post-staining

Abbreviations: UA - Uranyl acetate

Health & Safety requirements

Ensure you are familiar with all relevant local Health and Safety documents and procedures (Material Safety Data Sheets, Procedural Risk Assessments, Standard Operating Procedures etc.) before starting.

Safety critical: All procedures designated safety critical must be performed in a fume hood and the following PPE must be worn - lab coat, gloves, and eye protection.

Materials

- ☐ Balance
- ☐ Measuring cylinders
- ☐ 50 ml Falcon tube
- ☐ Aluminium foil
- ☐ Roller mixer
- ☐ Parafilm Agar Scientific AGG3398
- ☐ Fridge, 4°C

Chemicals

- ☐ ddH₂O; double distilled water
- ☐ Uranyl acetate [UO₂(CH₃COO)₂] Agar Scientific AGR1260A

Procedure

Note: "UA is one of the heaviest stains used in electron microscopy, it can be used as both a positive and negative stain. Uranyl ions combine in large amounts with nucleic acids via their phosphate and carboxyl groups (not by sulphhydryl groups). Treatment with UA following fixation with osmium tetroxide and glutaraldehyde improves the appearance of membranous and other structures. UA completely penetrates the entire thickness of a thin section. When a grid is floated, section facing downwards, the UA solution permeates all the cellular components throughout the thickness of the section" ^[1].

Preparation of 1% aqueous uranyl acetate

1. Zero a balance with an empty 50 ml tube.
2. Transfer tube to a fume hood, then add approximately 0.5 g uranyl acetate powder to the tube. **Safety critical**

Uranyl Acetate	State: Solid	Hazard Codes: H300, H330, H373, H411	
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3. Close the tube, remove from fume hood, and weigh it on the balance. **Safety critical**

Note: Do not take an open container of uranyl acetate outside the fume hood.

4. Return the tube to the fume hood and add enough water to make a 1% solution. For example, for 0.46 g of uranyl acetate, use 46 ml of water. Close the tube. **Safety critical**
5. Mix the solution using a vortex for 1-2 min.
6. Wrap the tube in aluminium foil to protect from light and place the tube on a rolling mixer for 3 hours, vortexing every hour until the uranyl acetate is dissolved.
7. Store at 4°C.

Note: Filter 1% uranyl acetate solution before use.

References:

[1] Hyatt, M.A. (2000) *Principles and Techniques of Electron Microscopy Biological Applications (Fourth Edition)*. Cambridge University Press. 342-350.